

Solar Charger EMI Filter Design Choices and Values

Circuit Theory:

To achieve an EMI filter with a smooth result and a long lifetime, we must account for a few things. We must select a type of filter structure that has a sharp falloff to ensure proper noise reduction, aim to keep the output voltage as stable as possible, make sure there is surge protection within the filter, and ensure there is no resonance. In short, the filter must be able to sustain itself for an extended period of time without repair or maintenance.

To achieve an effective amount of noise reduction, I have decided to use a standard pi-filter, also known as a C-L-C filter, due to its structure of an inductor between two capacitors in parallel. This will provide us with two stages of attenuation. The first capacitor eliminates any high-frequency noise; the inductor will block differential noise that makes it past the first capacitor; and the last capacitor should eliminate any remaining noise. Because the pi-filter is a third-order filter, the signal roll-off will be sharper, resulting in more dramatic noise removal. Lastly, to account for coupling and to further remove noise, we will place capacitors with small values alongside the main ones.

Generally, to ensure the stability of the input voltage and to obtain a small DC resistance, the inductor must be as small as possible. This allows us to freely select a capacitor value, and then account for the inductor last.

To protect the components in the filter from the effects of power fluctuations due to obstruction between the filter and power supply, a TVS diode rated for our voltage amount should be placed at the start of the circuit. This will eliminate any unexpected spikes and protect the components for an extended period of time.

Lastly, we must account for any resonance within the circuit. While this isn't as large an issue compared to the others, adding a resistor in series after the inductor will ensure this won't become an issue in the future.

Value Selection:

To start with, the solar charger has a maximum input voltage of 19.29V and operates at a frequency of 1MHz. This means that we want the cutoff frequency to be below our operating point of 1MHz; usually, we would select it to be a decade below our cutoff point, which is calculated below.

$$f_c = \frac{f_{sw}}{10} = \frac{1MHz}{10} = 100kHz$$

Now, since we have a pi-filter, we use both capacitors and inductors, which both have their own reactances. To simplify, the reactance of a capacitor, X_C , acts like a high-pass filter, and the reactance of an inductor, X_L , acts like a low-pass filter. We can manipulate these values to cut the high and low noise separately and create an area in which our regular frequency can pass through normally. We assume the reactance X_C to be at a value of 1, to which we can then derive a value for capacitance.

$$X_C = \frac{1}{2\pi C f_c} \Rightarrow C = \frac{1}{2\pi X_C f_c}$$

$$C = \frac{1}{2\pi * 100kHz * 1\Omega} \Rightarrow C = \frac{1}{628318.5307} \Rightarrow C = 1.592\mu F$$

This means that at each node we will need a capacitance of around 1.6μF to have a reactance of 1Ω at 100kHz. But by adding a small 10μF to each capacitor, we can eliminate the low-end frequency as well. Doing this will give us a new reactance value.

$$X_c = \frac{1}{2\pi * 100kHz * 10.1\mu F} \Rightarrow X_c = \frac{1}{6.34601716} \Rightarrow X_c = 0.158\Omega$$

This is well below the initial estimate for reactance, which means that the circuit is doing a good job at cutting out noise.

As for the second capacitor, we won't place as large a capacitor because most of the noise is already being filtered out. So I will select a common capacitor size, but small enough to ensure we are below our initial reactance estimate of 1Ω at 100kHz.

$$X_c = \frac{1}{2\pi * 100kHz * 4.71\mu F} \Rightarrow X_c = \frac{1}{2.95938028} \Rightarrow X_c = 0.338\Omega$$

Again, the value is well below the expected amount. All that's left from here is to calculate the inductance based on the capacitor values we have selected.

$$f_c = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{(2\pi f_c)^2 C} \Rightarrow L = \frac{1}{(2\pi * 100kHz)^2 * 10\mu F}$$

$$L = \frac{1}{3947841.76} \Rightarrow L = 25.33nH$$

This value is the theoretical minimum value we can place for our specific circuit. While this may work, it is preferable to round up the value to give us some room for error when the signal attenuates. For this reason, I have selected our inductance value to be 4.7μH, as not only is

it larger than what we need, giving us that space, but it is a common value for a real-world inductor.

Given the fact that I have chosen different values than those calculated for our components, the actual cutoff frequency is changed.

$$f_c = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{4.7\mu H * 10.1\mu F}} = \frac{1}{4.32901892 * 10^{-5}} \cong 23.1kHz$$

Just to confirm how much our filter cuts off, we can calculate attenuation to make sure it fits our needs.

$$Signal\ Attenuation = 40 \log_{10}\left(\frac{f_{sw}}{f_c}\right) = 40 \log_{10}\left(\frac{1MHz}{23.1kHz}\right) = 40 * 1.636 \cong 65dB$$

This means that the noise at 1MHz is reduced by a factor of almost 200.